RR-B-600E
February 1, 1971
SUPERSEDING
Fed. Spec. RR-B-600D
October 26, 1966
(See 6.5)

FEDERAL SPECIFICATION

BOTTLES, VACUUM

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers vacuum insulated bottles.

1.2 Classification.

1.2.1 Types, grades, styles, and sizes. The vacuum bottles shall be of the following types, grades, styles and sizes as specified, (see 6.2).

Type I - Replaceable 1-quart glass filler (inner shell).

Grade A - Terne or black-plate outer shell.
Grade B - Carbon-steel outer shell.

Type II - Corrosion-resisting steel inner shell.

Grade A - Corrosion-resisting steel outer shell, narrow mouth.

Style a - Without spigot.

Size - 1-quart.
Size - 2-quart.

Style b - With spigot, 2-quart.

Grade B - Carbon steel outer shell, wide mouth, without spigot, 1 quart.

FSC 7330

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2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-390	-	Plastic Molding Material, Polyethylene, Low and Medium Density.
L-P-398	-	Plastic Molding Material, Styrene-Butadiene.
QQ-A-250/1	-	Aluminum Alloy 1100, Plate and Sheet.
QQ-B-613	-	Brass, Leaded and Non-Leaded: Flat Products (Plate, Bar, Sheet and Strip).
QQ-B-626	-	Brass, Leaded and Non-Leaded; Rod, Shapes, Forgings, and Flat Products with Finished Edges (Bars, Flat Wire, and Strips.)
QQ-S-698	-	Steel, Sheet and Strip, Low-Carbon.
TT-E-489	-	Enamel, Alkyd, Gloss (for Exterior and Interior Surfaces).
TT-P-636	-	Primer Coating, Alkyd, Wood and Ferrous Metal.
ZZ-S-751	-	Stoppers, Bottle, Rubber.
LLL-S-731	-	Stoppers, Bottle, Cork.
PPP-B-636	-	Box, Fiberboard.
PPP-B-843	-	Cushioning Material, Cellulosic.

Federal Standards:

Fed. Std. No. 123	-	Marking for Domestic Shipment (Civil Agencies).
Fed. Std. No. 595	-	Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers, at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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Military Specifications:

MIL-R-3065 - Rubber, Fabricated Parts.
MIL-P-17091 - Polyamide (Nylon), Plastic, Rigid; Molded Parts, Rods and Flats.
MIL-P-22748 - Plastic, Polyethylene, and Copolymers, High Density, Molding and Extrusion Material.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads, 40" x 48" Pallets, Skids, Runners, or Pallet Type Base.
MIL-STD-417 - Rubber Compositions, Vulcanized General Purpose, Solid (Symbols and Tests).

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 2146 - Polypropylene Molding and Extrusion Materials (Tentative).
A167-63 - Corrosion-Resisting Chromium Steel Plate, Sheet and Strip.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202 Union Station, 516 W. Jackson Blvd., Chicago, Illinois 60606.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., Agent 1616 P Street, N.W., Washingtion, DC 20036.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

3. REQUIREMENTS

3.1 First article. This specification contains provisions for first article test and approval (see 4.2, 6.2, and 6.4).

3.2 Standard product. The vacuum bottles delivered under this specification shall be the manufacturer's standard current commercial product except for any changes necessary to comply with specification requirements.

3.3 Materials. Materials not definitely specification shall be of the quality normally used by the manufacturer for his vacuum bottles, provided the completed bottles comply with all provisions of this specification. Unless otherwise specified herein, commercial tolerances shall apply to metal thicknesses.

3.3.1 Steel, carbon, sheet and strip. Carbon steel sheet and strip shall conform to cold-rolled, commercial quality of QQ-S-698.

3.3.2 Steel, corrosion-resisting, sheet and strip. Corrosion-resisting steel sheet and strip shall conform to the requirements for type 302, 304, or 305 of ASTM A167-63.

3.3.3 Aluminum sheet. Sheet aluminum shall conform to the requirements for alloy 1100 temper 0 of QQ-A-250/1.

3.3.4 Brass. Brass sheet for forming or drawing shall conform to composition 2 of QQ-B-613. Brass for machining shall conform to composition 24 of QQ-B-613 or composition 22 of QQ-B-626.

3.3.5 Plastics.

3.3.5.1 Polypropylene, virgin. The polypropylene shall conform to type I-00400 of ASTM D 2146.

3.3.5.2 Styrene, butadiene. Styrene butadiene shall conform to type III, class 2 of L-P-398.

3.3.5.3 Polyethylene virgin. Virgin polyethylene shall conform to type I, grade 2 or 3 of L-P-390.

3.3.5.4 Polyethylene, high density. High density linear type virgin polyethylene shall conform to type I, grade I of MIL-P-22748.

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3.3.5.5 Polyamide. Polyamide plastic shall conform to type I of MIL-P-17091. The color shall be black.

3.3.5.6 Acrylonitrile butadiene styrene (ABS). ABS material shall have Federal Drug Administration (FDA) approval for use in contact with food.

3.3.6 Rubber.

3.3.6.1 Rubber, synthetic. Synthetic rubber shall be a rubber composition meeting the requirements of type, class, grade, and suffix letters RS-310A_1B of MIL-STD-417 and MIL-R-3065.

3.3.6.2 Stoppers, rubber. Rubber stoppers shall conform to type A, grade I of ZZ-S-751.

3.3.7 Stoppers, cork. Cork stoppers shall conform to grade C of LLL-S-731.

3.3.8 Primer. Primer shall conform to TT-P-636.

3.3.9 Enamel. Enamel shall conform to class B of TT-E-489. Color shall be black No. 17038 of FED-STD-595.

3.4 Construction. Details not specifically defined hereinafter shall be in accordance with the manufacturer's standard practice, provided the completed bottles comply with all provisions of this specification.

3.4.1 Type I vacuum bottle. The type I vacuum bottles shall be grade A or B as stated in 1.2.1 and shall consist basically of an outer shell constructed of metal; a vacuum type, narrow mouth, 1-quart capacity, replaceable glass inner shell; a stopper; a metal or plastic shoulder; and a screw-on cap, useable as a cup. When the glass replaceable inner shell is loaded from the bottom of the outer shell, the collar or shoulder shall be an integral part of the outer shell and the bottom cap shall be threaded to the outer shell. The glass inner shell shall be held in place at its base by either a fixed or movable shock absorber and at its neck by a rubber or plastic seal between the liner and shoulder of the outer shell. Water having a temperature between 203°F. and 207°F. in the vacuum bottles shall not decrease below 135°F. after 24 hours, when tested in accordance with 4.4.1 and 4.4.1.1. The capacity of the type I vacuum bottles shall be within 2 ounces of its rated capacity when tested in accordance with 4.4.2.

3.4.1.1 Outer shell, type I bottles. The outer shell of the type I, grade A bottle shall be fabricated from terne plates or black plate. The outer shell of the type I, grade B bottle shall be fabricated of the carbon steel specified in 3.3.1. The outer shells shall be internally coated to prevent rust, corrosion, and discoloration. When the collar or shoulder is not an integral part of the container, the outer shell shall be prepared to receive the collar or shoulder specified in 3.4.1.5.

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3.4.1.2 Inner shell, type I bottles. The replaceable glass inner shell shall be double-walled construction and formed of clear lime or borosilicate glass fully annealed. The inside surface shall be smooth and free from seeds, pits, stones, or striae. The chambered surface shall be silvered and the chamber evacuated of air.

3.4.1.3 Stopper and pouring lip, type I bottles. Stoppers and pouring lips shall be made of plastic conforming to 3.3.5.1, 3.3.5.2, 3.3.5.3, or a combination of these materials. Rubber specified in 3.3.6.2 shall be an acceptable alternate material for stoppers.

3.4.1.4 Screw-on cap or cup, type I bottles. A screw-on cap made of aluminum specified in 3.3.3 or plastic specified in 3.3.5.1 or 3.3.5.2, shall be provided with internal threads for attachment to the bottle outer shell or shoulder. The screw-on cap shall be suitable for use as a cup.

3.4.1.5 Collar or shoulder, type I bottle. The collar or shoulder when not an integral part of the outer shell, shall be fabricated of plastic specified in 3.3.5.1 or aluminum specified in 3.3.3, threaded for attachment to the outer shell, or permanently attached by roll forming or crimping operations.

3.4.1.6 Collar, rubber, type I bottle. When a shock absorbing rubber collar is required, it shall be made of rubber specified in 3.3.6.1.

3.4.1.7 Shock-absorber base, type I bottle. The base shock absorber shall be molded or formed from plastic specified in 3.3.5.2 or 3.3.5.4, rubber specified in 3.3.6.1, or at the option of the manufacturer may be a steel spring type.

3.4.2 Type II vacuum bottles. The type II vacuum bottles shall be grade A, style a, size 1 or 2-quart; or grade A, style b, size 2-quart; or grade B, without spigot, size 1-quart, as stated in 1.2.1. All type II bottles shall have inner shells of corrosion-resisting steel specified in 3.3.2. The grade A bottle shall have a narrow mouth. The grade B bottle shall have a wide mouth. The inner and outer shells shall be joined by welding or brazing, and the space between the shells shall be insulated by vacuum or by a combination of vacuum and insulating material. When filled with water between 203°F. and 207°F. the water shall not fall below the temperature shown in table I when tested in accordance with 4.4.1 and 4.4.1.2. The capacities of the bottles shall be as shown in table I when tested in accordance with 4.4.2.

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TABLE I. Type II Vacuum Bottles: Capacities And Heat Retention

Grade	Style	Size (± 2 oz.)	Minimum Temperature (Degrees F)
A	a	1 quart	170
A	a	2 quart	172
A	b	2 quart	160
B	-	1 quart	145

3.4.2.1 Grade A outer shell. The type II, grade A, outer shell shall be fabricated of corrosion-resisting steel specified in 3.3.2.

3.4.2.2 Grade B outer shell. The outer shell of the type II, grade B vacuum bottle except for the threaded cup seat shall be fabricated of cold-rolled carbon steel conforming to 3.3.1. The shell shall be strengthened by two or more ribs, one rib and shoulder, stepping, or indenting, or any combination of these methods in accordance with the manufacturer's standard practice.

3.4.2.3 Threaded cup seat and shoulder. The threaded cup seat and shoulder of the grade B outer shell shall be fabricated of corrosion-resisting steel specified in 3.3.2.

3.4.2.4 Bottom cap.

3.4.2.4.1 Grade A. The bottom cap for the grade A vacuum bottles shall be made of corrosion-resisting steel specified in 3.3.2.

3.4.2.4.2 Grade B. The bottom cap for the grade B vacuum bottle shall be fabricated from either carbon steel conforming to 3.3.1 or corrosion-resisting steel specified in 3.3.2.

3.4.2.5 Stoppers.

3.4.2.5.1 Narrow mouth bottles. A stopper made of cork specified in 3.3.7, or plastic conforming to 3.3.5.1, 3.3.5.2, 3.3.5.3, or a combination of these plastics shall be provided for narrow mouth vacuum bottles except style b (see 3.4.2.5.2).

3.4.2.5.2 Bottles with spigots and bottles with wide mouths. The type II, grade A, style b bottle and the type II, grade B wide mouth bottle shall have stoppers of the rubber specified in 3.3.6.1 or 3.3.6.2, the plastic specified in 3.3.5.1, 3.3.5.2, or 3.3.5.3, or a combination of these plastics.

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3.4.2.6 Caps. All type II vacuum bottles shall be provided with a screw-on cap. The cap for style a shall be usable as a cup. The caps shall be fabricated from any of the following:

(a) Brass specified in 3.3.4.
(b) Corrosion-resisting steel specified in 3.3.2.
(c) Plastic specified in 3.3.5.1.
(d) Plastic specified in 3.3.5.1 or 3.3.5.4, acrylonitrile butadiene styrene conforming to 3.3.5.6, or any combination of these plastics, with a metal overcap. The overcap shall be corrosion-resisting steel, or chrome plated brass or carbon steel. When solid plastic cups are furnished, the wall thickness of the plastic cup shall be not less than 0.090 inch.

3.4.2.7 Bail. The type II, grade B bottle, shall be provided with a fold-down bail affixed to the cap specified in 3.4.2.6. The bail shall be fabricated of brass or of corrosion-resisting steel. The bail shall be securely attached to the posts so that it cannot be easily disengaged. There shall be no permanent deformation or rupture to the bail when subjected to a 20 pound test load in accordance with 4.4.3.

3.4.2.7.1 Bail posts. The bail post shall be fabricated of brass conforming to 3.3.4 or corrosion-resisting steel conforming to 3.3.2 not less than 0.312 inch wide. The posts shall be attached to the cap in accordance with the manufacturer's standard practice.

3.4.2.8 Inner shells, grades A and B. Inner shells for grades A and B shall be fabricated of corrosion-resisting steel specified in 3.3.2.

3.4.2.9 Spigot assembly. The type II, grade A, style b vacuum bottle shall be provided with a spigot and vent tube. The vent tube shall extend to within 1/2 inch of the bottom of the inner shell and shall be used as an air bleed, so that the vacuum bottle may be emptied when inverted. When tested in accordance with 4.4.4, vacuum bottles with spigot assembly shall not leak in excess of 4 ounces of water per hour or become deformed when subjected to 15 pounds per square inch gage (p.s.i.g.) internal pressure. In addition, when tested at atmospheric pressure for one hour, in accordance with 4.4.4, there shall be no evidence of leaks.

3.4.2.9.1 Spigot. The spigot assembly shall be fitted with a tee-type spigot and except for the handle shall be fabricated of corrosion-resisting steel or brass, chromium plated. The handle shall be molded of plastic specified in 3.3.5.5. The spigot shall be made to incorporate an air bleed and a cock to open and close the vent tube at the same time as the liquid exits.

3.5 Finish.

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3.5.1 Type I. The shells of the type I bottles shall be lithographed or enameled in accordance with the manufacturer's standard practice.

3.5.2 Type II.

3.5.2.1 Grade A. The type II, grade A bottles shall have a 2B or better finish on the interior surface of the inner shell. The exterior of the outer shell and the exterior of the bottom caps shall have a 2B or better finish, a dull frosted finish, or a brushed No. 4 commercial polish.

3.5.2.2 Grade B. The type II, grade B bottles shall have a 2B or better finish on the interior surface of the inner shell. The carbon steel outer shells shall be primed with one coat of the primer specified in 3.3.8 and finished with one coat of the enamel specified in 3.3.9 except that carbon steel bottom caps shall be nickel-chromium plated.

3.5.3 Caps. Corrosion-resisting steel caps shall have a 2B or better, or a dull frosted finish on all surfaces. Brass caps shall have a decorative nickel-chromium finish on all surfaces or a decorative nickel-chromium finish on the exterior surface and a tin coated or plated inner surface.

3.5.4 Bail and bail posts. Brass bail and bail posts shall be chromium plated. Corrosion-resisting steel bail and bail posts shall be finished in accordance with the manufacturers standard practice.

3.6 Marking for identification. Each vacuum bottle shall be permanently and legibly marked with the manufacturer's name, tradename, or trademark, easily identifiable with such manufacturer.

3.7 Workmanship. The finished vacuum bottles shall be clean and free of slivers, die marks, sharp edges, dents, scratches, and deformities.

3.7.1 Welding and brazing. Welding and brazing shall be uniform, sound and free of burn holes, cracks, undercutting, excessive metal build-up, and splatter. All scale and flux shall be removed and the welds or brazes shall be finished to blend with the adjacent joined metal surfaces.

3.7.2 Rivets. Rivets shall be tight and rivet heads and peened ends shall be round and in full contact with the surfaces of the joined material. There shall be no empty rivet holes.

3.7.3 Enamel finish. The enamel finish shall be dry, complete and uniform in coverage and color and free of embedded foreign matter, smudges, finger marks, smear, orange peel, drips, sags, wrinkles, blisters and runs. There shall be no areas of poor or no coverage or any visible evidence of rust showing through the finish coat.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 First article inspection. When a preproduction sample is required, it shall be examined for the defects in table II and for compliance with dimensions specified and tested in accordance with the applicable tests in table III. Failure to pass the applicable tests shall be cause for rejection of the preproduction sample. The presence of any of the defects listed in table II or any dimension not within specified requirements shall be cause for rejection of the preproduction sample.

4.3 Inspection. Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 Components and materials inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 In process inspections.

4.3.2.1 Internal coating, type I bottles. Examination shall be made of the outer shells of the type I grade A and grade B bottles to determine compliance with the internal coating requirements of 3.4.1.1. Whenever nonconformance is noted, correction shall be made to the affected items and the lot in-process.

4.3.2.2 Primer coat, type II, grade B bottle. Examination shall be made of the outer shell of the type II, grade B bottle to determine compliance with the primer coat requirements of 3.5.2.2. Whenever nonconformance is noted, correction shall be made to the affected items and the lot in-process.

4.3.3 End item inspection. The lot shall consist of all vacuum bottles of one type, grade, size, and style offered for inspection at one time. The sample unit shall be one complete vacuum bottle.

4.3.3.1 Visual examination. Examination shall be made of the vacuum bottles for the defects listed in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE II. Classification of defects

Examine	Defect	Classification Major	Minor
Finish, exterior, Type I, grades A and B and type II,	Not lithographed or enamel finish, not dry, poor or not coverage, visible rust, incomplete coverage.	X	
	Not uniform in color or coverage, smudges, embedded foreign matter, finger marks, smears, orange peel, drips, sags, runs, wrinkles, or blisters.		X
Type II, grade A	Not a 2B finish or better on interior of inner shell. Exterior of outer shell and exterior of bottom caps not finished as specified.	X	
Construction, Type I	Inner shell not replaceable.	X	
	Inner shell not glass double wall vacuum type.	X	
	Cap not of aluminum or plastic.	X	
	Cap not screw-on type, not usable as a cup.	X	
	Inner shell not held in place at base by fixed or removable rubber or plastic shock absorbers, or steel spring.	X	
	Neck of inner shell not held in place by rubber or plastic seal.	X	
	Visible seeds pits, stones, striae, or scratches on inside of glass filler.	X	
	Chambered surfaces of glass filler not silvered.	X	
	Stoppers not of plastic or rubber.	X	
	Pouring lip not of plastic.	X	
Type I, grade B	Outer shell not of carbon steel.	X	
Construction, Type II	Inner shell cup seat and shoulder not of corrosion-resisting steel.	X	
	Inner and outer shells not joined by welding or brazing.	X	
	Cup seat and shoulder not threaded or not corrosion-resisting steel.	X	
Type II, grade A	Outer shell and bottom cap not of corrosion-resisting steel.	X	

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification	
		Major	Minor
Construction (cont'd)			
Type II, grade B	Outer shell not of carbon steel.	X	
	Outer shell not strengthened by two or more ribs, or one rib and shoulder, stepping or indenting or a combination of these methods.	X	
	Bottom cap not of corrosion-resisting steel or carbon steel.	X	
	Bail missing.	X	
	Bail not fold down type.		X
	Bail posts or bails not of brass or corrosion-resisting steel.	X	
	Overcaps, when required, missing or not chrome plated brass or carbon steel or corrosion-resisting steel.	X	
Type II, style a	Stopper missing.	X	
	Stopper not of plastic or cork (narrow mouth).	X	
Type II, style b and grade B	Stopper missing.	X	
	Stopper not of rubber or of plastic.	X	
Type II, grades A and B	Cap missing.	X	
	Cap not usable as a cup.		X
	Cap not screw-on type.	X	
	Cap not of brass, corrosion-resisting steel, or plastic or plastic liner with metal overcap.	X	
Type II, grade A Style b	Spigot missing.	X	
	Vent tube missing.	X	
	Spigot not tee type.		X
	Spigot body not of chrome plated brass or of corrosion-resisting steel.	X	
	Spigot does not have an air bleed and a cock to open and close vent when handle is activated.	X	
	Spigot handle not of molded plastic.		X
Workmanship	Dirty, die marks, or scratches.		X
	Slivers, sharp edges, dents, or deformities.	X	

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TABLE II. Classification of defects (con.)

Examine	Defect	Classification	
		Major	Minor
Welds and brazes	Unsound, undercut, burn holes.	X	
	Excessive metal build up, splatter, flux and scale not removed, not finished to blend with adjacent surfaces.		X
Rivets	Loose, missing, not in full contact.	X	
	Rivet heads or peened ends not round.		X
Marking for identification	Missing, illegible, incorrect, or incomplete.		X

4.3.3.2 Dimensional examination. An examination shall be made to determine compliance with dimensions specified. Any nonconformance shall be considered a defect. The inspection level shall be S-2 with an AQL of 4.0 expressed in terms of defects per hundred units.

4.3.4 Testing of the end item. The end item shall be tested as specified in Table III.

TABLE III. End item test requirements

Classification of bottles to be tested	Test	Inspection level	Test paragraph	Requirement paragraph
All Type I	Heat retention	S-3	4.4.1 and 4.4.1.1	3.4.1
All Type I	Capacity	S-2	4.4.2	3.4.1
All Type II	Heat retention	S-3	4.4.1 and 4.4.1.2	3.4.2 & Table I
All Type II	Capacity	S-2	4.4.2	3.4.2 & Table I
Type II, Grade B	Bail deformation	S-2	4.4.3	3.4.2.7
Type II, style b	Spigot leakage	S-2	4.4.4	3.4.2.9

4.3.5 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking as required by section 5 of this specification are complied with. Defects shall be as specified in table IV The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5 defects, expressed in terms of defects per hundred units.

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TABLE IV. Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect, illegible; improper size, location, sequence, or method of application.
Materials	Component missing or damaged.
Contents	Weight of contents of shipping containers exceeds requirements.

4.3.5.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with PPP-B-636.

4.3.5.2 Examination of palletization. An examination shall be made to determine that palletization is in compliance with section 5 of this specification. Defects shall be as specified in table V. The sample unit for this examination shall be one palletized unit load ready for shipment. The lot shall be the number of palletized unit loads submitted for inspection at one time. The inspection level shall be S-1 with an AQL of 6.5 expressed in terms of defects per hundred units.

TABLE V. Examination of palletization

Examination	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement.
Palletization	Not as specified. Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with straps specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect, illegible, of improper size, location, sequence, or method of application.

4.4 Tests.

4.4.1 Heat retention test. Heat retention tests shall be conducted at least 3 days after the drawing of vacuum and sealing of the vacuum chamber.

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4.4.1.1 Type I vacuum bottles. The type I bottles shall be filled to the neck with water at a temperature between 203°F and 207°F, and the stopper secured. The bottle shall then be stored for not less than 24 hours in an ambient temperature between 68°F and 72°F. Following the required storage time, the temperature of the contents shall be taken to determine compliance with 3.4.1. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.1.2 Type II vacuum bottles. The completed type II bottles shall be filled with water at a temperature between 203°F and 207°F., and the stopper secured. Following not less than 6 hours storage in an ambient temperature between 68°F and 72°F., the temperature of the contents shall be taken to determine compliance with 3.4.2 and table I. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.2 Capacity. The capacity shall be determined by filling the bottle to stopper height with water poured from a graduated measure, to determine compliance with 3.4.1 and table I. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.3 Bail deformation test. The assembled bail and fittings of the type II, grade B vacuum bottle shall be tested by suspending a weight of 20 pounds from the bail. The vacuum bottle shall be secured rigidly and the weight suspended from the center of the bail so as to develop the greatest stress on the fittings. The bail and fitting shall be examined for compliance with 3.4.2.7. Any nonconformance with specified requirements shall constitute failure of this test.

4.4.4 Spigot-leakage test. Type II, style b bottles shall be filled to 1/2 capacity with clean water. The bottle shall then be subjected to an internal pressure of not less than 15 p.s.i. The spigot shall then be closed to retain the internal pressure. The bottle shall then be inverted for not less than one hour and any resulting leaks measured to determine compliance with 3.4.2.9. The internal pressure shall then be stabilized at atmospheric pressure with the bottle 1/2 filled with clean water and inverted for one hour, following which there shall be no leaks. Any nonconformance with specified requirements shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each bottle shall be packaged in a close-fitting corrugated fiberboard box conforming to style RSC (end-opening), type CF, variety optional class domestic of PPP-B-636. Type I bottles shall be immobilized and cushioned by means of scored and folded fiberboard pads or die-cut sheets or be completely wrapped in cellulosic cushioning material conforming to type II, class B of PPP-C-843. Closure shall be in accordance with the appendix of PPP-B-636.

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5.1.2 Level C. Bottles shall be packaged to afford adequate protection against damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice, when it meets this requirement.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. Bottles of one type, grade, style, and size only, packaged as specified in 5.1, shall be packed in a close-fitting fiberboard shipping container conforming to style RSC, V2s of PPP-B-636. Closure, waterproofing by means of tape, and reinforcing with flat strapping or tape banding shall be in accordance with the appendix of PPP-B-636. The weight of content of each shipping container shall not exceed 65 pounds.

5.2.2 Level B. Bottles of one type, grade, style, and size only, packaged as specified in 5.1 shall be packed in a close-fitting fiberboard shipping container conforming to class domestic of PPP-B-636. Closure shall be in accordance with method II of the appendix of PPP-B-636. The weight of contents of each shipping container shall not exceed 65 pounds.

5.2.2.1 When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s, fabricated in accordance with PPP-B-636 and closed in accordance with the appendix thereto.

5.2.3 Level C. Bottles, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification or National Motor Freight Classification, as applicable.

5.3 Palletization, (Military requirements). Unless otherwise specified (see 6.2), bottles, packed as specified in 5.2, shall be palletized in accordance with Load Type I of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking.

5.4.1 Civil agencies. In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with FED-STD-123. Precautionary marking requirements shall apply to type I bottles.

5.4.2 Military requirements. In addition to any special marking required by the contract or order, interior boxes and shipping containers shall be marked in accordance with MIL-STD-129. Special handling marking requirements applicable to fragile items shall apply to type I bottles.

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6. NOTES

6.1 Intended use. The vacuum bottles covered by this specification are intended to be used to maintain contents at or near their original temperatures, for a predetermined period of time.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type, grade, style, and size vacuum bottle required (see 1.2.1).
(c) First article (see 3.1, 4.2, and 6.4).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
(e) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
(f) When palletization is not required (see 5.3).

6.3 Civil agency procurement. When level B packaging is required for civil agency procurements the level A requirements of 5.1.1 shall apply.

6.4 First article. The items covered by this specification require first article test and approval under appropriate provisions of paragraph 7-104.55 of the Armed Services Procurement Regulations. The first article should be a preproduction sample. The first article should consist of one unit. The contracting officer should include specific instruction in all procurement instruments, regarding arrangements for examination, test, and approval of the first article.

6.5 Classification comparison.

6.5.1 Type I. The classification "Class 1, Narrow mouth" and "Size 1-quart" under type I, grade A of superseded RR-B-600d have been deleted as all type I bottles are narrow mouth and 1-quart size.

6.5.2 Type II. The classification "Class 1, Narrow mouth" under type II, grade A of superseded RR-B-600d has been deleted as all type II, grade A bottles are narrow mouth. The classifications "Class 2 - Wide mouth," "Style a - Without spigot," and "Size - 1-quart" under type II, grade B - Carbon steel outer shell of superseded RR-B-600d have been deleted as all type II, grade B bottles are wide mouth, without spigot, size 1-quart.

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Army - MD
Navy - MC, MS

Preparing activity:

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CIVIL AGENCIES INTEREST:

GSA-FSS
HEW
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION Bottles, Vacuum RR-B-600E

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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